

Harmful Brain Activity Classification using EEG Signals

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1st Shruti Kamble

Computer Engineering Department,
VIIT, Pune
Pune, India
shrutikamble3032@gmail.com

2nd Dhanashri Mahale

Computer Engineering Department,
VIIT, Pune
Pune, India
dhanashri.22220088@viit.ac.in

3rd Vedant Vedpathak

Computer Engineering Department,
VIIT, Pune
Pune, India
vedant.22110246@viit.ac.in

4th Pratik Rathod

Computer Engineering Department,
VIIT, Pune
Pune, India
pratik.22110066@viit.ac.in

5th Yogesh Sharma

Computer Engineering Department,
VIIT, Pune
Pune, India
yogesh.sharma@viit.ac.in

Abstract—The accurate classification of harmful brain activity is crucial in medical contexts, such as epilepsy monitoring units and brain-computer interface (BCI) systems. In this study, we focus on developing robust algorithms to automatically detect and classify harmful brain events using electroencephalography (EEG) data. Our goal is to enhance diagnostic accuracy, reduce false positives, and enable faster intervention. We examined machine learning models to detect seizures and other pathological states using EEG recordings, such as XGboost, Support Vector Machines (SVM), KNN, Random Forests, and Decision Trees. We evaluate our approach on two datasets—the DEAP dataset (originally designed for emotion analysis) and the HMS-Harmful Brain Activity Classification dataset from Kaggle. Our proposed model achieves promising results, with 99.2% accuracy on the DEAP dataset and an impressive 96% accuracy on the HMS dataset using XGboost. Our work contributes to advancing medical diagnostics and improving patient care by automating the detection of harmful brain activity from EEG signals.

Index Terms—Harmful Brain Activity Classification, Machine learning, XGboost, SVM, Random Forest, Decision tree

I. INTRODUCTION

The human brain is a complex network of interconnected neurons that communicate through electrical impulses. Electroencephalography (EEG) is a valuable tool for monitoring brain activity by recording these electrical signals from the scalp. Clinicians and researchers use EEG data to diagnose various neurological conditions, including epilepsy, sleep disorders, and brain injuries.

Detecting harmful brain activity, such as epileptic seizures, abnormal patterns, or other pathological events, is pivotal for timely intervention and effective treatment. Conventional methods often employ visual inspection of EEG traces by experts, it might be subjective and time-consuming. As a

result, there's increased curiosity about creating automated algorithms to classify EEG data and identify harmful brain activity.

In this research, we focus on developing robust algorithms for classifying harmful brain activity using EEG data. Our goal is to enhance diagnostic accuracy, reduce false positives, and enable faster intervention. Specifically, we explore machine learning models to automatically detect and classify different types of harmful brain activity. We evaluate our approach on two distinct datasets: the DEAP dataset and the HMS-Harmful Brain Activity Classification dataset from Kaggle

The paper is structured as follows:

- I. Introduction
- II. Dataset Used
- III. Problem Description
- IV. Literature Review
- V. Proposed Methodology
- VI. Result and Analysis
- VII. Conclusion and Future work

II. DATASET USED:

A. DEAP Dataset:

The dataset DEAP (Database for Emotion Analysis using Physiological Signals) has EEG recordings from multiple subjects during emotional stimuli. It includes labeled segments corresponding to various emotional states, such as calm, happy, sad, and fearful. Despite being intended for emotion analysis, DEAP's EEG data records relevant neural patterns, such frequency bands, which may also signal detrimental brain activity, such as seizures. To identify these pathological states, transfer learning may be used to modify features from DEAP. Thus, by using prior neural patterns for expanded applications, DEAP helps overcome data limits.

B. HMS-Harmful Brain Activity Classification Dataset :

The HMS-Harmful Brain Activity Classification dataset, sourced from Kaggle, provides EEG recordings specifically from patients with known harmful brain activity. Annotations within this dataset indicate the presence or absence of harmful events, allowing us to train and evaluate our models effectively. This dataset is particularly valuable for our research, as it directly addresses our classification task.

III. PROBLEM DESCRIPTION

The central problem addressed in this research is the accurate classification of harmful brain activity using EEG signals. Specifically, we aim to detect and categorize seizures and other from EEG recordings. Our goal is twofold: (1) to improve the efficiency of existing diagnostic methods by automating the process, and (2) to enhance patient outcomes by providing timely interventions based on reliable predictions. By leveraging machine learning models, we intend to achieve high accuracy in identifying harmful brain activity, ultimately benefiting patients, clinicians, and researchers alike.

IV. LITERATURE REVIEW

[1]Using physiological measurements and subjective assessments of valence and arousal, the study attempts to categorize six fundamental emotions-anger, disgust, fear, pleasure, sorrow, and surprise-into two classes of symmetric categories: emotion and no emotion. The EEG data and subjective emotional assessments were taken from the DEAP dataset, which was used by the researchers. The following were the maximum categorization accuracy values for each emotion: Joy, Surprise, Disgust, Fear, sadness and anger are 100%, 96%, 95%, 90.75%, 90.08%, and 98.02% respectively. The study shows how well machine learning methods work to classify valence emotions based on EEG readings.

[2]The goal of the study was to predict epileptic seizures with enough time to spare before the actual seizure started. They have a better true positive rate with their model. They took advantage of the CHB-MIT dataset, which contains EEG recordings taken from epileptic sufferers. For preprocessing, empirical mode decomposition, or EMD, was utilized. In order to train the prediction model, characteristics from the time and frequency domains were retrieved. Before a seizure began, the average prediction time was 23.6 minutes; the maximum prediction time recorded was 33.46 minutes; the average sensitivity was 92.23%; the average specificity was 93.38%. enhancing the preprocessing of EEG signals even more for greater sensitivity. investigating adaptive window widths and hybrid preprocessing techniques. creating a real-time seizure prediction web system.

[3]With a particular focus on the pre-ictal stage (prior to the start of a seizure), the researchers sought to anticipate epileptic seizures. The suggested method forecasts when epileptic episodes will occur. It functions in three states: Before the seizure begins: the pre-ictal stage. Ictal state: At the time of the convulsion. After the seizure, in the post-ictal state. The program predicts the pre-ictal state by evaluating

EEG data, giving an average lead time of about 34 minutes prior to the actual seizure. 34 minutes is the average pre-ictal time. Medical professionals can give medication to prevent seizures during this lead time. 88.90% of people had seizures during the ictal state, according to the true positive rate (TPR).

[4]Predicting epileptic seizures is the aim; this can greatly enhance patient safety and quality of life. The first step of the technique is to use EEG time series to create an embedding space. The samples with the highest information are then chosen using an improved Poincaré plane tailored to the data. On the Poincaré plane, 64 fuzzy rules are created in each channel to characterize small dynamics. The distribution of these ambiguous rules' frequencies within each minute is used to extract features. Higher variance characteristics are designated as ictal features, and Principal Component Analysis (PCA) is used to further reduce these features. The approach uses Support Vector Machines (SVM) to assess the change from interictal to preictal state. An average sensitivity of 91.8% was reached by the algorithm during testing on 460 hours of EEG data from 19 patients.

"TABLE. I" shows summary of EEG Signal Classification results from different research.

TABLE I
SUMMARY OF EEG SIGNAL CLASSIFICATION RESULTS

Ref. No.	Dataset	Classifier	Metric
1	DEAP	Random Forest	Accuracy (98.2)
2	MIT DATABASE	SVM	Sensitivity (92.23)
3	MIT DATABASE	KNN	Sensitivity (97.44)
4	Freiburg Database	SVM	Sensitivity (96.6)
5	EPILEPSIAE database	SVM	Sensitivity (73.98)
6	DEAP	CNN	AUC (96.71)
7	BETA SEED DepEEG	EEGFormer EEGFormer EEGFormer	BETA-Accuracy (57.29) SEED-Accuracy (80.12) DepEEG-Accuracy (60.12)
8	DEAP	SNNs	Accuracy (96.67)

V. PROPOSED METHODOLOGY

A. System Architecture

An overview of the technologies and methodologies used in this study is provided in this section. This section provides an overview of general procedure, as shown in "Fig. 1". The whole process mainly includes preprocessing of the data, training of the model, assessment, and comparative analysis. Firstly, In particular, feature extraction, normalization, and cleansing are part of the data preprocessing step are performed on both the datasets DEAP and Harmful Brain Activity. Following that, machine learning algorithms are trained and optimized, such as Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Decision Tree, Random Forest, and XGBoost. To evaluate the performance of the model, evaluation metrics like F1-score, accuracy, precision, and recall are used. Finally, an inter-dataset comparative study is carried out to identify the optimal algorithm for identifying detrimental brain activity.

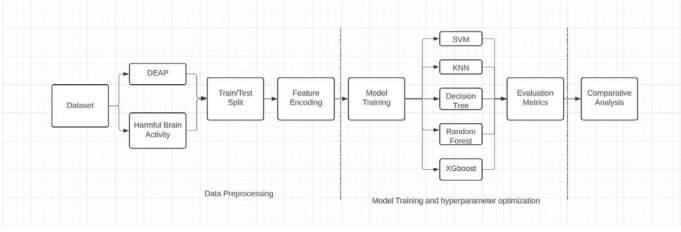


Fig. 1. Architecture of the System

B. Models

We explored several machine learning techniques to classify harmful brain activity:

- **XGBoost:** XGBoost is a gradient boosting algorithm known for its high performance and robustness. Several weak learners (decision trees) are combined to produce a powerful ensemble model. We fine-tune hyperparameters and train an XGBoost classifier on our EEG data.
- **Support Vector Machines (SVM):** Binary classification problems are well suited for Support Vector Machines (SVM). SVMs identify the ideal hyperplane in the feature space for dividing various classes. We experiment with different kernel functions (linear, radial basis function, etc.) to find the best SVM model.
- **Random Forests:** Multiple decision trees are aggregated using ensemble models called random forests. A majority vote determines the final forecast, with each tree having been trained on a random portion of the data. Random Forests are interpretable and handle noisy data well.
- **Decision Trees:** Decision Trees are simple yet interpretable classifiers. They split the feature space based on specific thresholds, creating a tree-like structure. Although prone to overfitting, decision trees provide insights into feature importance.

VI. RESULTS AND ANALYSIS

A. Model Performance Metrics:

Our best performing XGboost model's performance is shown in "Fig. 2" and "Fig. 3" on Harmful Brain Activity dataset.

B. Harmful Brain Activity and DEAP dataset accuracy metric:

"Fig. 4" and "Fig. 5" shows the accuracies of machine learning models on HMS and DEAP dataset.

"TABLE. II" demonstrates how various models perform on EEG datasets. The table indicates that the XGBoost model outperforms the other models in terms of accuracy. It has 99.20% accuracy on DEAP and 96.61% accuracy on Harmful Brain Activity Classification dataset.

C. Confusion Matrix Analysis:

A confusion matrix is used to provide a more thorough analysis. This matrix provides insights into how well the algorithm handles both classes by graphically displaying the distribution of correct and incorrect predictions. The "Fig. 6" shows confusion matrix of our model.

Classification Report:				
	precision	recall	f1-score	support
0	0.97	1.00	0.98	14719
1	0.96	0.71	0.82	1358
2	0.92	0.76	0.83	468
3	0.96	0.99	0.98	3937
4	0.93	0.88	0.90	338
5	0.97	0.78	0.86	348
6	0.96	0.95	0.95	55
7	0.97	0.89	0.93	64
8	0.88	0.74	0.80	19
9	0.91	0.67	0.77	15
10	0.60	0.60	0.60	10
11	0.75	0.75	0.75	4
12	1.00	0.33	0.50	3
13	0.86	0.75	0.80	8
14	1.00	1.00	1.00	3
15	1.00	0.50	0.67	10
17	0.00	0.00	0.00	1
accuracy			0.97	21360
macro avg	0.86	0.72	0.77	21360
weighted avg	0.97	0.97	0.96	21360

Fig. 2. Classification Report

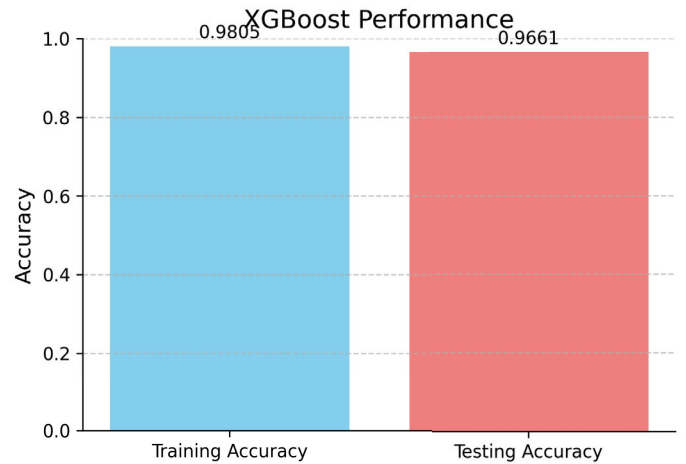


Fig. 3. XGboost Performance

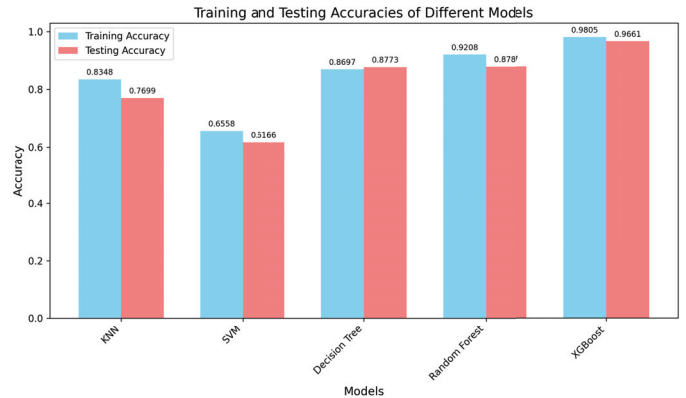


Fig. 4. HMS dataset

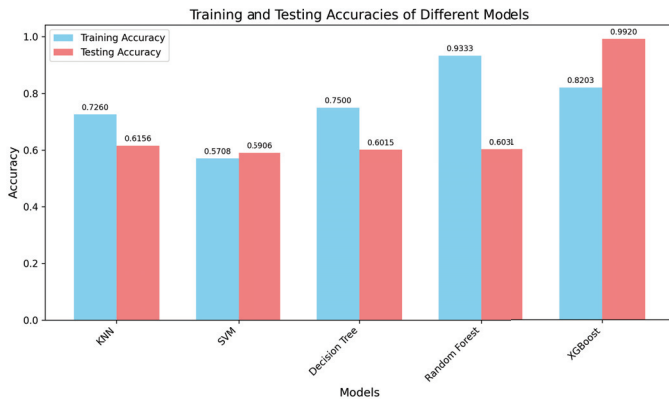


Fig. 5. DEAP dataset

TABLE II
PERFORMANCE OF DIFFERENT MODELS ON EEG DATASETS

Dataset	Model	Training Acc.	Testing Acc.
DEAP	KNN	0.7260	0.6156
	SVM	0.5708	0.5906
	Decision Tree	0.7500	0.6015
	Random Forest	0.9333	0.6031
	XGBoost	0.8203	0.9920
Harmful Brain Activity	KNN	0.8348	0.7699
	SVM	0.6558	0.6166
	Decision Tree	0.8697	0.8773
	Random Forest	0.9212	0.8792
	XGBoost	0.9803	0.9661

D. False Positive Rate (FPR) Analysis

The “Fig. 7” shows FPR of our model. Each class has a different False Positive Rate (FPR), where lower rates correspond to greater model performance. For example, Classes 1, 5, and 10 have high FPR, indicating frequent misclassifications, whereas Classes 0 and 12 have low FPR, indicating good classification accuracy. These findings emphasize the need for focused enhancements to raise diagnostic accuracy and lower false positive rates.

VII. CONCLUSION AND FUTURE WORK

In this study, we addressed the critical task of classifying harmful brain activity using electroencephalography (EEG) data. Our proposed approach leveraged machine learning models, including XGboost, SVM, Random Forests, and Decision

False Positive Rate (FPR) for class 0: 0.0370
False Positive Rate (FPR) for class 1: 9.7500
False Positive Rate (FPR) for class 2: 3.6452
False Positive Rate (FPR) for class 3: 0.3289
False Positive Rate (FPR) for class 4: 1.9091
False Positive Rate (FPR) for class 5: 9.6250
False Positive Rate (FPR) for class 6: 1.5000
False Positive Rate (FPR) for class 7: 3.5000
False Positive Rate (FPR) for class 8: 2.5000
False Positive Rate (FPR) for class 9: 5.0000
False Positive Rate (FPR) for class 10: 1.0000
False Positive Rate (FPR) for class 11: 1.0000
False Positive Rate (FPR) for class 12: 0.0000
False Positive Rate (FPR) for class 13: 2.0000
False Positive Rate (FPR) for class 14: 0.0000
False Positive Rate (FPR) for class 15: 0.0000
False Positive Rate (FPR) for class 16: 0.0000

Fig. 7. False Positive Rate

Trees, to automatically detect and categorize seizures and other pathological events. With 99.2% accuracy on the DEAP dataset and a remarkable 96% accuracy on the HMS dataset using XGboost, the findings showed promising accuracy.

Although our current research does not include distributed system design, adding distributed processing could greatly enhance our model. Distributed systems can improve the capability of managing large amounts of data and support real-time processing as the model grows, especially in clinical situations that demand fast and precise analysis.

Future studies should investigate incorporating security measures to protect sensitive EEG data and utilizing distributed system methods to improve scalability and real-time processing capabilities. Furthermore, it will be essential to focus on specific enhancements to decrease the number of false positives in order to improve diagnostic precision. By concentrating on these areas, we can improve the practical usefulness of our model and help improve patient care and diagnostic practices.

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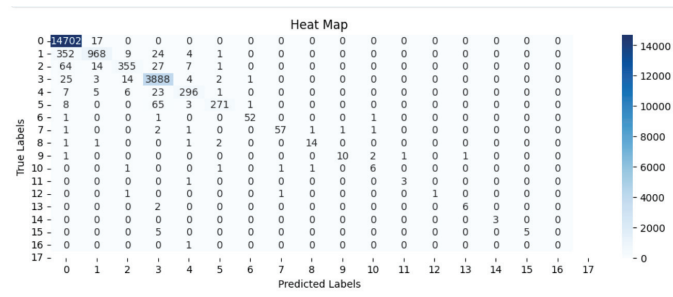


Fig. 6. Confusion Matrix

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